Understanding Object Hallucination in Vision-Language Models

1. Introduction

Vision-language models (VLMs) can produce fluent descriptions and answers that are not fully supported by the image. We focus on one specific failure mode: object hallucination, where a model claims that an object is present even though it is absent under the chosen ground-truth annotation.

For example, an image containing a dog and frisbee may produce “dog, frisbee, tennis ball”; the tennis ball is hallucinated. Missing an object is an omission, calling a dog a cat is a recognition error, changing a red car to blue is an attribute error, and reversing a spatial relation is a relation error. Keeping these categories separate lets us ask a more useful question: when visual evidence becomes uncertain or incomplete, what does a VLM rely on when generating an answer?

We are interested in empirical understanding rather than architecture design. We can vary visual and linguistic conditions, compare publicly available models, and examine whether consistent patterns emerge across benchmarks and model families.

2. Objective

We investigate when and why object hallucination occurs in open vision-language models, with particular attention to visual evidence, scene complexity, language priors, object frequency and co-occurrence, prompt formulation, and model characteristics. The study can be broad or focused: a single carefully controlled hypothesis is sufficient, while multiple complementary experiments can be used when they reveal different aspects of the phenomenon.

3. Scope, Models and Non-Goals

The study can use publicly released, reproducible models at different levels of visual-language capability. Examples include ViT and DeiT as vision-only controls; CLIP for image-text alignment; BLIP and BLIP-2 for multimodal understanding/generation; and LLaVA or comparable open VLMs for instruction-following generation. Other open models may be included where their checkpoints and evaluation procedure are reproducible.

We do not need to build a new foundation model or pursue state-of-the-art performance. Lightweight inference-time interventions, prompting, controlled perturbations, representation analysis, or small auxiliary components are appropriate when they help answer the research question.

4. Research Questions

- How frequently do different open VLMs hallucinate objects, and how does the measured rate depend on the evaluation protocol?
- How does hallucination change with scene complexity, object count, object size, salience, position, or occlusion?
- When visual evidence is degraded or ambiguous, do models increasingly rely on object frequency, co-occurrence, or other language priors?
- How sensitive is hallucination to prompt or question formulation?

- Do different model families or scales exhibit different hallucination profiles?
- Do hallucination, omission, recognition, attribute, and relation errors change together or trade off under harder visual conditions?

- Do conclusions from binary object probing agree with conclusions from free-form generation?

5. Experimental Setting

We can construct controlled comparisons around any factor that is scientifically interesting. For example, the same image can be evaluated before and after blur, compression, resizing, occlusion, or other visual changes; scenes can be compared by object count or composition; object size and position can be varied; common and uncommon objects can be contrasted; and typical versus atypical object co-occurrences can be examined. Prompt wording, question type, decoding settings, model family, and model scale can likewise be varied. The important principle is to make the intended factor identifiable and keep other relevant conditions as comparable as possible.

We can also combine naturally occurring benchmark examples with controlled perturbations. This allows a finding observed on a benchmark to be checked under a more targeted experimental condition rather than assuming that correlation alone explains the behaviour.

6. Benchmarks

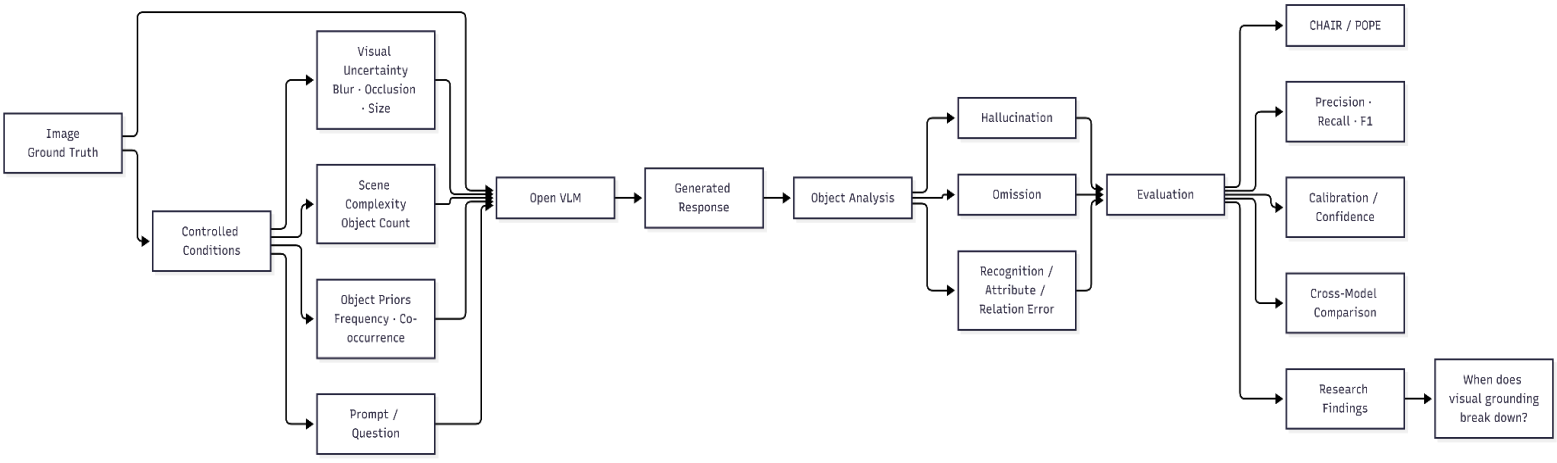
POPE is a natural primary benchmark for object hallucination. It frames object existence as controlled yes/no probing and includes random, popular, and adversarial negative sampling, making it useful for studying false-positive behaviour and contextual priors.

CHAIR on MS COCO is useful for free-form captioning because it measures hallucinated object mentions against image annotations. It provides a complementary view to binary probing.

THRONE extends object-hallucination evaluation to free-form VLM generations and can be used when the research question concerns generated responses rather than isolated object queries.

HallusionBench is broader than object hallucination, but can be useful for examining whether an observed grounding failure extends to visual illusion or language-driven errors.

Additional benchmarks can include MME, MM-Vet, SEED-Bench, NoCaps, GQA, and A-OKVQA where they support a particular hypothesis or provide a useful secondary check. They should remain complementary rather than becoming a benchmark-coverage exercise.



7. Evaluation Metrics

- The metric set can be chosen according to the research question. For object hallucination, CHAIRs and CHAIRi can quantify sentence-level and object-mention-level hallucination in captions, while POPE precision, recall, and F1 can characterize false-positive and false-negative behaviour in object probing. Hallucination rate can also be reported when its denominator is explicitly defined.
- Complementary metrics can help distinguish hallucination from simply saying less. Object precision/recall, omission rate, per-class error rates, and confusion patterns can reveal whether a reduction in hallucination is accompanied by increased omissions or substitutions. Where relevant, attribute and spatial-relation accuracy can be added. Caption-quality metrics such as CIDEr or SPICE may be useful as secondary measures, but should not substitute for hallucination-specific metrics.
- If usable confidence or probability information is available, we can examine calibration and uncertainty through confidence distributions, reliability curves, ECE, Brier score, or the ability of confidence to separate correct from hallucinated responses. We can report quantities such as over-confidence frequency, confidence conditional on error type, and changes in confidence under visual degradation. These analyses are optional and should only be used when the model's confidence signal has a defensible interpretation.
- Results can be summarized with effect sizes, confidence intervals, distributions, per-condition quantities, and paired comparisons. The emphasis should be on the magnitude and consistency of observed effects rather than on statistical significance alone. Multiple comparisons should be handled transparently when many exploratory conditions are tested.

8. Ablation Studies

Ablations can be used to isolate the source of an observed effect. Useful comparisons include random versus popular versus adversarial negatives in POPE; single-object versus multi-object scenes; original versus degraded images; visible versus occluded objects; common versus uncommon objects; typical versus atypical co-occurrences; alternative prompt formulations; free-form versus binary evaluation; and different model families or scales. We can select the ablations that directly test the explanation being proposed rather than applying every ablation by default.

9. Deliverables

- A concise research report defining the phenomenon, research questions, experimental setting, findings, and limitations.
- Reproducible code and configuration covering the principal experiments, prompts, model checkpoints, and evaluation procedure.
- Quantitative results with hallucination-specific and relevant complementary metrics, supported by plots or tables.
- Controlled or ablation experiments that test the main explanation being investigated.
- Qualitative examples showing representative hallucination, omission, recognition, attribute, or relation failures.
- A final discussion distinguishing observed evidence from interpretation and identifying unresolved questions.

10. References

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- Yu et al. (2024), “MM-Vet: Evaluating Large Multimodal Models for Integrated Capabilities,” ICML 2024.
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- Li et al. (2022), “BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation,” ICML 2022.
- Li et al. (2023), “BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models,” ICML 2023.
- Alayrac et al. (2022), “Flamingo: a Visual Language Model for Few-Shot Learning,” NeurIPS 2022 — relevant multimodal foundation-model background.

Related work that may also inform experimental design includes studies of visual grounding, multimodal compositionality, visual contrastive decoding, uncertainty and calibration, and language-prior effects in VLMs. The references above provide a compact starting set; additional literature can be selected according to the specific hypothesis investigated.